

Low Resistance and High Stable Solid–Liquid Electrolyte Interphases Enable High-Voltage Solid-State Lithium Metal Batteries

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Solid-state batteries (SSBs) with addition of liquid electrolytes are considered to possibly replace the current lithium-ion batteries (LIBs) because they combine the advantages of benign interfacial contact and strong barriers for unwanted redox shuttles. However, solid electrolyte and liquid electrolyte are generally (electro)-chemically incompatible and the resistance of the newly formed solid–liquid electrolyte interphase (SLEI) appears as an additional contribution to the overall battery resistance. Herein, a boron, fluorine-donating liquid electrolyte (B, F-LE) is introduced into the interface between the high-voltage cathode and ultrathin composite solid electrolyte (CSE), which is fabricated by adhering a high content of nanosized $\text{Li}_{6.4}\text{La}_3\text{Zr}_{1.4}\text{Ta}_{0.6}\text{O}_{12}$ (LLZTO) with poly(vinylidene fluoride-co-hexafluoropropylene) (PVDF-HFP), to generate a low resistance and high stable SLEI in situ, giving a stable high-voltage output with a reinforced cathode|CSE interface. B, F-LE, consisting of a highly fluorinated electrolyte with a lithium bis(oxalato)borate additive, exhibits good chemical compatibility with CSE and enables rapid and uniform transportation of Li^+ , with its electrochemically and chemically stable interface for high-voltage cathode. Eventually, the B, F-LE assisted $\text{LiNi}_{0.6}\text{Co}_{0.2}\text{Mn}_{0.2}\text{O}_2|\text{Li}$ battery displays the enhanced rate capability and high voltage cycling stability. The findings provide an interfacial engineering strategy to turn SLEI from a “real culprit” into the “savior” that may pave a brand-new way to manipulate SLEI chemistry in hybrid solid–liquid devices.

1. Introduction

Electrolyte is media conveying ions and its properties greatly affect the performances of electrochemical energy storage devices.^[1,2] Hitherto, the vast majority of commercially available

lithium-ion batteries (LIBs) and next generation systems such as Li–S and Li–O₂ rely on liquid electrolytes (LEs) because they endow the exchanged Li^+ with fast bulk and interface kinetics.^[3–7] However, many insurmountable challenges arose from the use of LEs, such as their inherent leaking and flammability, diffusion of electrode-decomposition products in high-voltage LIBs, creating detrimental shuttle effect in Li–S (polysulfide) and Li–O₂ (redox mediators) batteries, restrict their widespread application in practical.^[8,9] Ultimately, solid-state batteries (SSBs) that rely on solid electrolytes (SEs; e.g., polymer and inorganic solid electrolyte) are considered to be effective in eliminating the thorny issues involved with using LEs.^[10–14] However, SSBs have been limited to obtain higher specific/volume energy density induced by some intrinsic flaws of SEs, including excessive thickness and weight, sluggish transport, and poor interface kinetics.^[15–20]

Neither batteries only relied on LEs, nor SSBs created a single solution for achieving the competitive energy densities so far.^[14] Most straight-

forward tactic is to fabricate hybrid solid–liquid batteries (anode|LE_{anolyte}|SE|LE_{catholyte}|cathode).^[21–26] Hybrid solid–liquid electrolyte concurrently contribute to benign interfacial contact and strong barrier that hinders the “cross-talk” of the electrodes, which is superior to the SEs with insufficient contact and poor interface kinetics, hence filling the gap in SSBs for ideal capacity output.^[21,24,27–29] Nevertheless, recent works have suggested that the interface between SE and LE is chemically unstable and forms a metastable solid–liquid electrolyte interphase (SLEI).^[26,30–32] SLEI can form at virtually any contact between an SE and an LE, and its component originates from the decomposition reactions of solvents, conducting salts as well as SEs.^[14] These decomposition products at SE|LE interface appear as an additional contribution to the overall battery resistance, representing a major barrier to the implementation of hybrid solid–liquid batteries. Conspicuously, SLEI plays a significant role in regulating interfacial stability and Li^+ transportation, even profoundly affecting the morphology and structural stability of electrode.^[26,31–35] Unsatisfyingly, the (electro) chemical compatibility between SE and LE has usually been overlooked in hybrid solid–liquid systems.

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Herein, a boron (B), fluorine (F)-donating liquid electrolyte consisting of lithium bis(trifluoromethanesulfonyl) imide/lithium bis(oxalato)borate (LiTFSI:LiBOB = 0.95:0.05 by mol) dual salts in a mixture of fluoroethylene carbonate (FEC)/1,1,2,2-tetrafluoroethyl-2',2'-trifluoroethylether (HFE)/methyl(2,3,3,3-tetrafluoro-2-(2,2,3,4,4,4-hexafluoro-3-(perfluoropropoxy)butoxy)propyl) carbonate (PFEMC) (FEC:HFE:PFEMC = 1:1:4 by weight) is introduced into the interface between high-voltage cathode and nano- $\text{Li}_{6.4}\text{La}_3\text{Zr}_{1.4}\text{Ta}_{0.6}\text{O}_{12}$ /poly(vinylidene fluoride-co-hexafluoropropylene) (nano-LLZTO/PVDF-HFP) composite solid electrolytes (CSEs). Stacking nano-LLZTO particles together using PVDF-HFP as gap adjusting fillers are beneficial for achieving 25 μm ultrathin dense membrane. The addition of LiBOB in highly fluorinated solvents can reduce the high resistive LiF content in SLEI, which not only accelerates ion transport kinetics at the cathode–electrolyte interface but also reinforces interfacial thermodynamic stability. Meanwhile, highly fluorinated solvents furnished with large excess of F^- can catalyze the decomposition reaction of the LiBOB to $\text{Li}_x\text{BO}_y\text{F}_z$ ionic conducting, which further decreases interface diffusion resistance. The resulting low resistance and high stable SLEI enables excellent cycling

stability and rate capability of $\text{LiNi}_{0.6}\text{Co}_{0.2}\text{Mn}_{0.2}\text{O}_2$ (NCM622)||Li cell under high-voltage conditions. Further, postmortem analyses in combination with ex situ X-ray photoelectron spectroscopy (XPS), scanning electron microscopy-energy dispersive X-ray spectroscopy (SEM-EDS), and nuclear magnetic resonance (NMR) provide insights on mechanisms of the interfacial stability improvements, which will be of guiding significance on future research in high-voltage interfacial stability for SSBs.

2. Results and Discussion

2.1. Synthesis Procedures and Structure Characterizations

The detail synthesis process is schematically depicted in Figure 1a. In this work, we adopt nano-LLZTO with the apparent particle size of 60–100 nm as inorganic host material ascribing to its higher ionic conductivity and better dispersibility, superior to micro-LLZTO (Figure 1b,c; Figure S1, Supporting Information).^[36] X-ray diffraction (XRD) profile shows the structure of nano-LLZTO and confirms a typical cubic phase structure, which matches well with the standard pattern of the

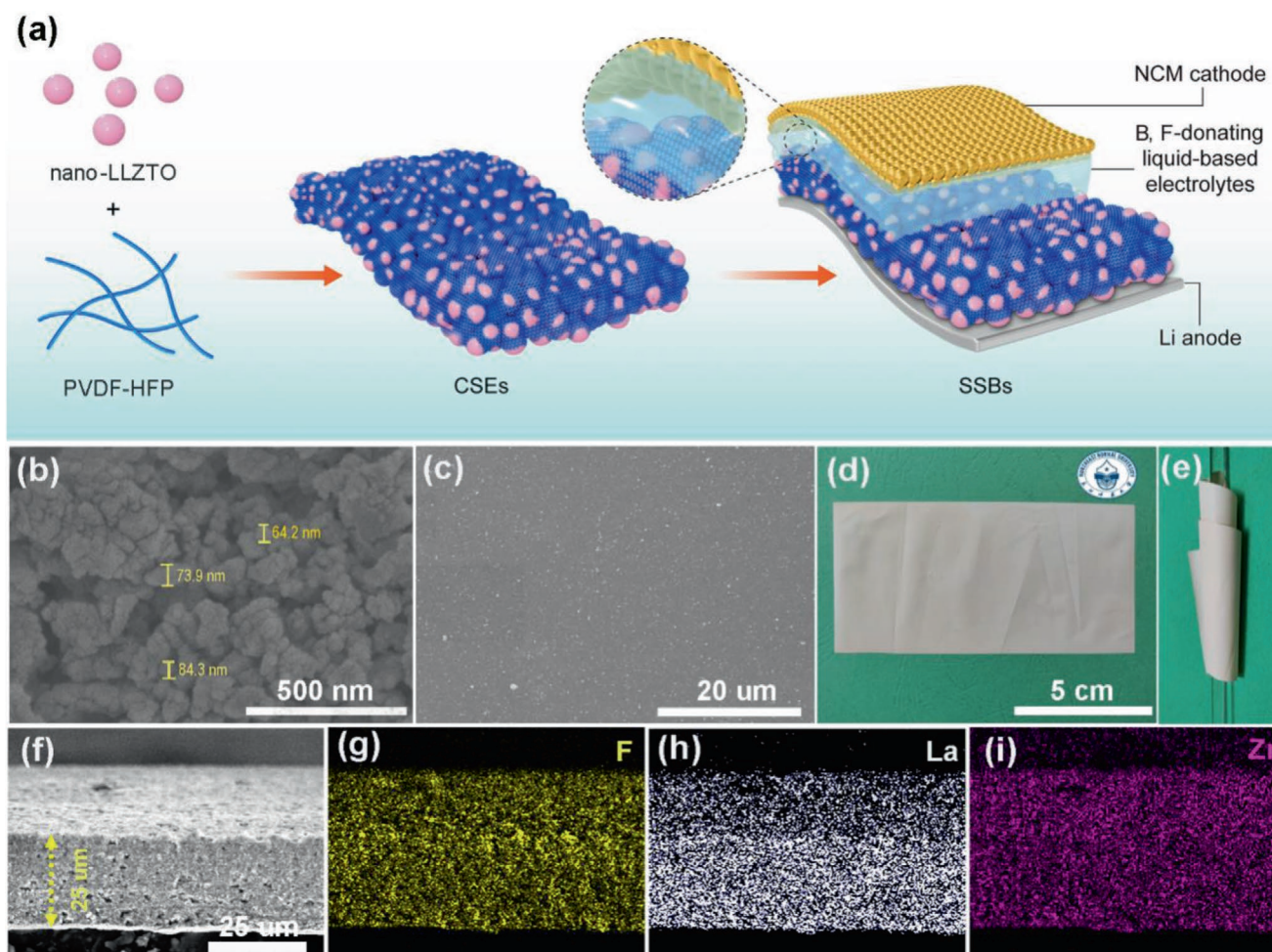


Figure 1. a) Schematic representation of the detail synthesis process of CSEs and the assembly process of SSBs. b) SEM image of the nano-LLZTO powders. c) SEM image of CSE. d,e) The apparent snapshots of CSE. f–i) SEM image and corresponding EDS mappings of CSE.

$\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ (PDF# 80-4057) (Figure S2, Supporting Information). PVDF-HFP due to its wide electrochemical window, higher mechanical strength, favorable thermal stability, and strong adhesive force is chosen as both organic matrix and binder to offset the gaps between inorganic particles, which then greatly enhances the wetting ability and reinforces the interfacial adhesion between electrode and SE membrane. Subsequently, the CSEs consisting of PVDF-HFP and nano-LLZTO are fabricated by a simple blade-casting method. The obtained membrane exhibits ultrathin characteristic and good flexibility (Figure 1d,e), endowing a huge potential for practical applications. Eventually, SSBs are assembled by infiltrating a trace amount of B, F-donating liquid electrolyte on cathode-side to gain a stable interface between CSEs and NCM622 cathodes.

To achieve the best ionic conductivity of PVDF-HFP with high content of nano-LLZTO, the CSEs with various nano-LLZTO contents are also designed. With the increment of mass percent of nano-LLZTO fillers, the ionic conductivities of CSEs at ambient temperature increase initially, and decrease sluggishly above 50 wt% (Figure S3, Supporting Information). Ultimately, PVDF-HFP with 50 wt% nano-LLZTO can reach the preferable ionic conductivity of $1.11 \times 10^{-4} \text{ S cm}^{-1}$ (Figure S3b, Supporting Information). The corresponding XRD result demonstrates that the diffraction profile displays the main characteristic peaks of nano-LLZTO, and no new peak can be founded, indicating an excellent chemical compatibility of nano-LLZTO with PVDF-HFP (Figure S2, Supporting Information). The cross-section SEM image presented in Figure 1f shows that PVDF-HFP with 50 wt% nano-LLZTO can still exhibit a compact and smooth surface, and the uniform membrane thickness is only 25 μm . Furthermore, the corresponding EDS mapping images of F, La, and Zr confirm the good dispersibility of nano-LLZTO in PVDF-HFP (Figure 1g–i). Taking both structural homogeneity and electrochemical kinetic behavior into consideration, PVDF-HFP with 50 wt% nano-LLZTO is chosen for further study, referring as CSE.

2.2. Chemical Compatibility of CSE with Liquid Electrolyte

For a systematic assessment of the chemical reactivity of B, F-donating liquid electrolyte toward CSE, B, F-donating liquid electrolyte assisted CSE (hereafter referred to as B, F-CSE) and its counterparts F-donating liquid electrolyte assisted CSE (here

after referred to as F-CSE) and conventional alkyl carbonate liquid electrolyte assisted CSE (here after referred to as C-CSE) were prepared. Figure S4 in the Supporting Information shows the snapshots of B, F-CSE, F-CSE, and C-CSE before and after immersing in various liquid electrolytes for 10 days. For all liquids, there is no apparent color change of each CSE. Their phase analyses are further explored by XRD and Fourier transform infrared (FTIR) spectroscopies (Figure 2). Before testing, the soaked CSE was thoroughly washed with alcohol and dried at 80 °C in vacuum to totally remove the any residual liquids. After soaking in conventional alkyl carbonate liquid electrolyte, the peak intensity of C-CSE is obviously weaker and broader (Figure 2a), which may be attributed to the chemical reactivity of LLZTO with liquid electrolyte.^[26,37] By contrast, the XRD patterns of B, F-CSE and F-CSE demonstrate that all the major diffraction peaks consistent with that of CSE, indicating the good chemical compatibilities of both liquid electrolytes with CSE (Figure 2a). Furthermore, the results of FTIR spectroscopies also confirm the trends observed in XRD results. In Figure 2b, the FTIR bands appeared at 755, 836, 880, 1072, 1400, and 1190 cm^{-1} represent the CF_2 bending and skeletal bending, CH rocking, amorphous phase belt, CF_2 stretching, and CH_2 deformation of PVDF-HFP, respectively.^[38,39] The bands found at 630 and 590 cm^{-1} are ascribed to the La–O and Zr–O stretching vibration from LLZTO. After immersing in B, F-donating and F-donating liquid electrolyte, the characteristic peaks of PVDF-HFP and LLZTO are still similar to those in unsoaked CSE, without any appearing or disappearing of the characteristic peaks. It indicates that the interactions between CSE and B, F-donating/F-donating liquid electrolytes are both simply physical mixtures without any chemical reactions. However, Figure 2b clearly reveals that besides the LLZTO and PVDF-HFP peaks, the peaks at 874 and 1500 cm^{-1} that can be assigned to Li_2CO_3 appear,^[26,40] implying the formation of Li_2CO_3 impurity phase after immersing in conventional alkyl carbonate liquid electrolyte.^[25,26]

2.3. Security and Electrochemical Compatibility of CSE with Liquid Electrolyte

Besides the good chemical compatibilities between solid and liquid electrolytes, combustion behaviors of hybrid solid–liquid

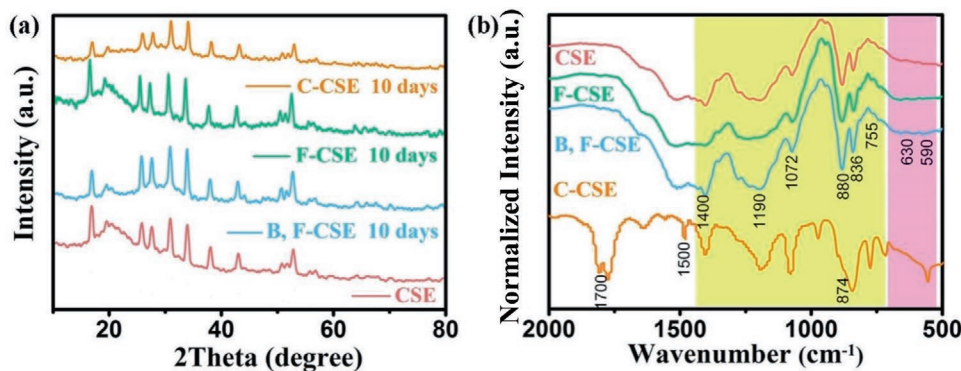


Figure 2. a) XRD of B, F-CSE, F-CSE, and C-CSE before and after immersing in various liquid electrolytes for 10 days. b) FTIR spectroscopies of B, F-CSE, F-CSE, C-CSE, and CSE.

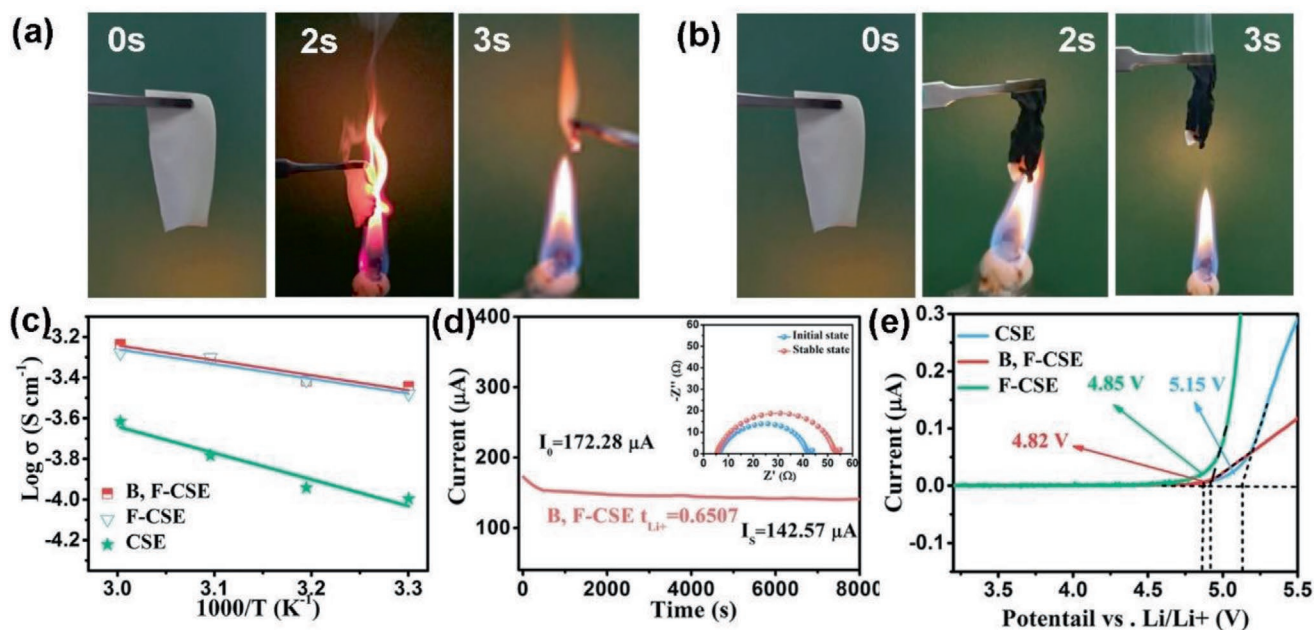


Figure 3. Burning tests of a) C-CSE and b) B, F-CSE. c) Arrhenius plots of B, F-CSE, F-CSE, and CSE. d) Current–time curve following a DC polarization of 20 mV of the Li|B, F-CSE|Li cell. Inset is the EIS plots before and after polarization. e) LSV curves of B, F-CSE, F-CSE, and CSE.

electrolytes are crucial for the securities of SSBs applications. Since the same solvent composition of B, F-donating and F-donating liquid electrolyte, hence we only compare the burning experiment results of B, F-CSE and C-CSE. As shown in Figure 3a, C-CSE can be easily ignited and burned continuously, which proves the potential safety hazard of hybrid solid–liquid electrolyte rely on alkyl carbonate liquid electrolyte. In comparison, B, F-CSE is merely charred and ashed without continuous flame (Figure 3b). This result is in good agreement with the burning results of perfluorinated or partially fluorinated electrolytes currently used in liquid-state LIBs. It can be reasonably explained by the theory that the fluorine substitution in the alkyl carbonates or ethers solvent could effectively suspend the propagation of oxygen radicals during combustion. In sum, the burning experiment results confirm the viability of using B, F-donating liquid electrolyte toward the development of SSBs with enhanced securities.

Ultiorly, the ionic transport properties of B, F-CSE were explored, which is of pivotal importance for potential practical application. The ionic conductivities of B, F-CSE, and its counterparts F-CSE and CSE are shown in Figure 3c. The CSE displays an ionic conductivity of $1.01 \times 10^{-4} \text{ S cm}^{-1}$ at 25°C , whereas the conductivity is dramatically enhanced with the addition of B, F-donating liquid electrolyte (B, F-CSE: $2.16 \times 10^{-4} \text{ S cm}^{-1}$) and F-donating liquid electrolyte (F-CSE: $2.10 \times 10^{-4} \text{ S cm}^{-1}$) compared with CSE. The improvement is thought to results from that the high ionic conductivities ($\sigma_{\text{B, F-donating liquid electrolyte}} = 6.16 \text{ mS cm}^{-2}$; $\sigma_{\text{F-donating liquid electrolyte}} = 6.07 \text{ mS cm}^{-2}$) and excellent wetting abilities of the liquid electrolyte penetrating the CSE, thus increasing more pathways for Li^+ transportation. Owing to the weakened Li^+ -solvation interaction by the fluorine-presence in the solvent molecules, the Li^+ transference numbers (t_{Li^+}) of B, F-CSE and F-CSE are 0.65 and 0.89, respectively, which are both higher than that of the CSE (0.394) (Figure 3d; Figure S5,

Supporting Information). The improvement of free Li^+ concentration in B, F-CSE and F-CSE electrolyte could reduce anion depletion and promote heterogeneous Li deposition. The electrochemical stabilities of B, F-CSE, its counterparts F-CSE and CSE were also recorded with linear sweeping voltammetry (LSV) curves in voltage range of open-circuit voltage to 5.5 V, as shown in Figure 3e. The oxidation decomposition potential of CSE is 5.15 V, benefiting from the inherent oxidation-tolerated characteristics of PVDF-HFP and LLZTO. By contrast, B, F-CSE and F-CSE exhibit relatively lower oxidation decomposition potentials of 4.82 and 4.85 V, separately. This result manifests that B, F-donating and F-donating liquid electrolyte have priority to be oxidized on the surface of active electrode under electrochemical situation, which would avoid the side reaction between cathode and CSE.

2.4. Li^+ Transportation Kinetics at Cathode|CSE Interfaces

To probe the interfacial compatibilities of B, F-CSE and its counterparts against high-voltage cathodes, the interfacial resistances of symmetric NCM622|NCM622 cells were investigated using electrochemical impedance spectroscopy (EIS) measurements at different time intervals, as exhibited in Figure 4a–c. Figure S6a in the Supporting Information shows the equivalent circuit used to fit the Nyquist plots and the fitting results are given in Table S1 in the Supporting Information. The EIS plot of NCM622|NCM622 symmetric cell shows a high frequency semicircle, which arises from the interfacial resistance R_{int} (cathode|liquid electrolyte ($R_{\text{C/LE}}$) + liquid|solid electrolyte ($R_{\text{S/LEI}}$)). The NCM622|NCM622 symmetric cell without liquid electrolyte presents a large interfacial resistance R_{int} (371 Ω) (Figure S6b, Supporting Information), whereas the NCM622|NCM622 symmetric cells employing C-CSE, F-CSE,

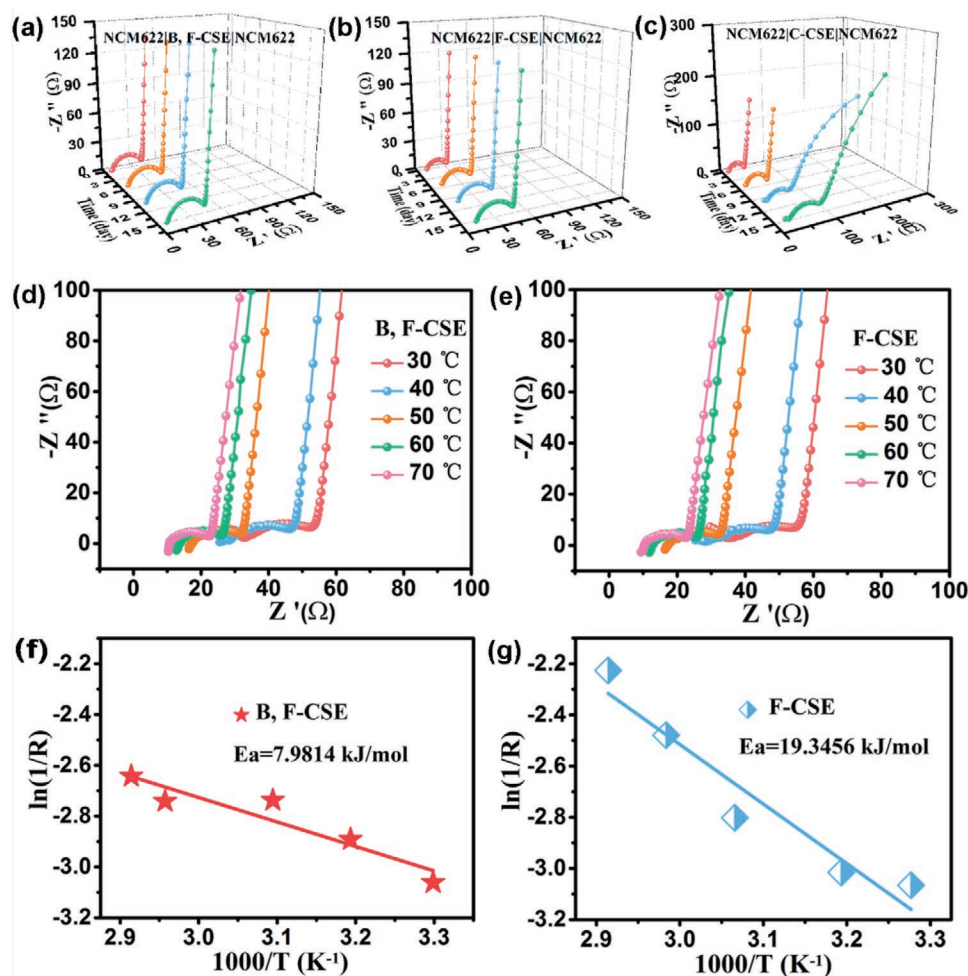


Figure 4. a–c) EIS plots of the NCM622|NCM622 symmetrical cells with B, F-CSE, F-CSE, and CSE at 25 °C as a function of the storage time. d,e) EIS plots of the NCM622|NCM622 symmetrical cells with B, F-CSE and F-CSE at different temperatures. f,g) Arrhenius behaviors of activation energies the R_{int} derived in EIS plots of the NCM622|NCM622 symmetrical cells with B, F-CSE and F-CSE.

and B, F-CSE display only 44.07, 22.16, and 30.96 Ω, respectively (Table S1, Supporting Information). These results clearly indicate a favorable wetting property of liquid electrolyte in reducing the interfacial resistance R_{int} . As the monitoring time increases, the interfacial resistance R_{int} between C-CSE and NCM622 cathode continuously grows from 44.07 to 1014.93 Ω over a period of 15 days. However, Figure S7 and Table S2 in the Supporting Information confirm that the interfacial resistances ($R_{\text{C/LE}}$) of cathode with various liquid electrolytes basically remain unchanged after 5 days. Therefore, the increasing interfacial resistance R_{int} may be specifically derived from the interfacial reaction of LLZTO with alkyl carbonate electrolytes, forming a high resistance SLEI with new phases, such as Li_2O , LiF , and Li_2CO_3 .^[26] In comparison, the NCM622|NCM622 symmetric cells with both B, F-CSE and F-CSE exhibit relatively low interfacial resistance R_{int} at the infancy stages, and their interfacial resistances R_{int} grow slowly along with the increase of time. Specifically, in the case of NCM622|NCM622 with B, F-CSEs, its interfacial resistance R_{int} becomes constant after 5 days. The temporal evolutions of interfacial resistances R_{int} between NCM622 and CSE assisting with various liquid electrolytes

illustrate their differences of kinetics at NCM622|CSE interfaces, indicating that the underlying mechanisms of B, F-CSE and F-CSE are different from that C-CSE.

Furthermore, the activation energies (E_a) for Li^+ transportation at NCM622|B, F-CSE and NCM622|F-CSE interfaces were evaluated, which were obtained based on the temperature-dependence of R_{int} derived in Nyquist plots of NCM622|NCM622 symmetric cell after monitoring for 15 days (Figure 4d,e). Both calculated E_a are in line with the law of Arrhenius. The E_a for Li^+ transfer at NCM622|B, F-CSEs interface is around 7.98 kJ mol⁻¹, smaller than that at NCM622|F-CSEs interface (19.35 kJ mol⁻¹) (Figure 4f,g). As is known to all, the Li^+ transfer across the interface is much more complex than Li^+ transportation in electrolyte. In a hybrid solid–liquid electrolyte, the activation energy barriers are not only influenced by solvation ability of solvent components in liquid electrolyte but also correlated with dynamical behavior for Li^+ transport through the SLEI. Since same solvent components and Li^+ concentration, B, F-donating and F-donating liquid electrolytes show a similar solvation structure, and we can infer that the composition and structure of SLEI have a dominant effect on

the interfacial Li^+ transfer kinetics. Further studies are probed by multiple postmortem analyses and illustrated in the following section.

2.5. Battery Performance under High-Voltage Conditions

To ensure a maximized energy density, in this part, NCM622 with a high active mass loading of 3.0 mg cm^{-2} and CSE with a thickness of $25 \mu\text{m}$ are used to evaluate the performance of as-prepared electrolyte in the high cut-off voltage electrochemical process. In the assembly process of full batteries, $2.5 \mu\text{L cm}^{-2}$ of various liquid electrolytes were dripped into NCM622/CSE interface. To exclude the impact of Li anode on electrochemical performance, the interfacial compatibility of Li anode with CSE was first explored in detail in the Supporting Information (Figures S8 and S9, Supporting Information). Figure 5a shows the initial charge–discharge curves for B, F-CSE, F-CSE, C-CSE, and CSE in NCM622/Li full batteries at a low rate of 0.1C (18 mA g^{-1}) in the voltage of $2.8\text{--}4.3 \text{ V}$. The battery with

dry CSE exhibits the initial charge capacity of 102.9 mAh g^{-1} , and discharge capacity of 72.1 mAh g^{-1} . The low Coulombic efficiency and strong polarization are mainly resulted from inferior physical contact of CSE with NCM622 cathode. Besides, the chemical reaction of LLZTO with the high-voltage cathode are usually accompanied at its interface with an electron source. The oxidation decomposition of LLZTO to undesirable phases (e.g., $\text{Li}_6\text{Zr}_2\text{O}_7$, Li_2O , Li_2CO_3 , $\text{La}_2\text{Zr}_2\text{O}_7$, and La_2O_3) occurs above 3.0 V , which gives rise to high electrochemical polarization.^[41] This high polarization and obvious capacity oscillation are also observed in the following cycles (Figure 5b). Comparing the NCM622/Li with CSE, NCM622/Li with F-CSE delivers a higher initial charge/discharge capacity of 163.8 and 157.5 mAh g^{-1} (Figure 5a). However, a continuous capacity decline for NCM622/Li with F-CSE is observed and the capacity decreases to $\approx 74\%$ in 50 cycles (Figure 5b). Conspicuously, with the synergistic effect of LiBOB additive and highly fluorinated solvents, the battery with B, F-CSE shows a high initial discharge capacity of 180 mAh g^{-1} at 0.1C , a high Coulombic efficiency of 74% , and a negligible polarization (Figure 5a). After 50 cycles,

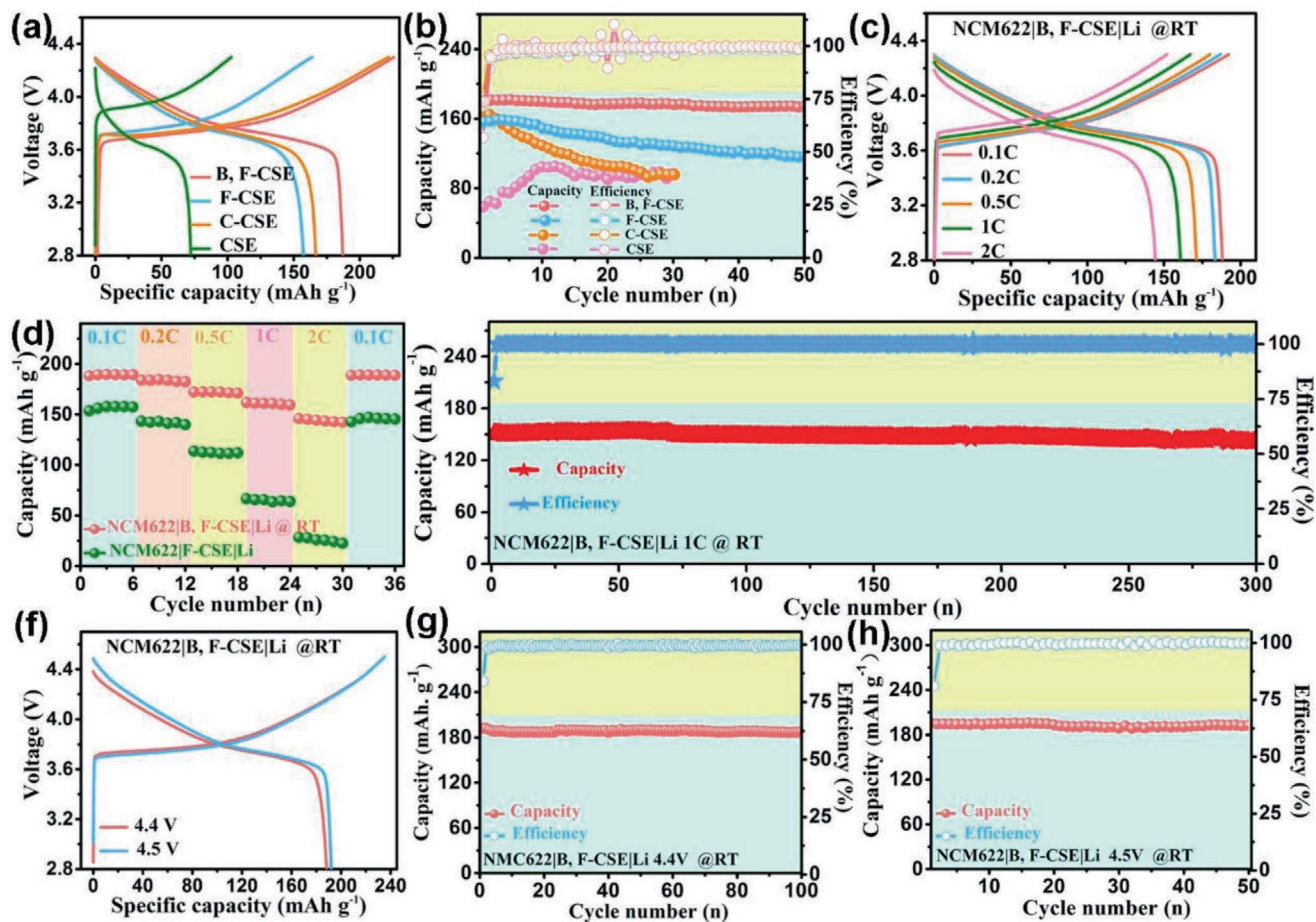


Figure 5. a) Voltage curves of the NCM622/Li cell with various electrolytes of the first cycle at 0.1C rate. b) Cycling performances of the NCM622/Li cell with various electrolytes at 0.1C rate. c) Voltage curves of the NCM622[B, F-CSE]/Li cell obtained at different rates from 0.1C to 2C . d) Discharge capacities versus cycle numbers of the NCM622[B, F-CSE]/Li cell cycled at different rates. e) Discharge capacity and Coulombic efficiency versus cycle number of the NCM622[B, F-CSE]/Li cell cycled at 1.0C . f) The initial voltage curves of the NCM622[B, F-CSE]/Li cell at 0.1C rate with the cutoff voltage of 4.4 and 4.5 V . g,h) Discharge capacity and Coulombic efficiency versus cycle number of the NCM622[B, F-CSE]/Li cell cycled at 0.1C with the cutoff voltage of 4.4 and 4.5 V . All the electrochemical characterizations were tested at ambient temperature.

the reversible capacity still retains 173.8 mAh g⁻¹ with a capacity retention of 96.5% (Figure 5b). Their improved cycling stabilities are also significantly better than that using conventional alkyl carbonate electrolyte (Figure 5b), underscoring the important role of liquid electrolyte component played in hybrid solid-liquid electrolytes.

The high-rate cycling performances of NCM622|Li with B, F-CSE and F-CSE were further evaluated at the rates from 0.1C to 2.0C. As shown in Figure 5c, the specific capacities at various rates clearly prove the better rate capabilities in NCM622|Li with B, F-CSE than that with F-CSE. The battery with B, F-CSE can reach a high capacity of 146 mAh g⁻¹ at a high rate of 2.0C, and the corresponding charge-discharge curves still present the weaker polarization, which is comparable to the performance of the conventional liquid-state LIBs (Figure 5d). The EIS results further verify the thermodynamics stability of NCM622|Li battery with B, F-CSE. As shown in Figure S10 in the Supporting Information, the battery resistance gradually increases in NCM622|F-CSE|Li over continuous time, while resistance in NCM622|B, F-CSE|Li remains a relatively stable within a 100 h period. It can be concluded that the lower battery resistance for NCM622|B, F-CSE|Li relative to that of the NCM622|F-CSE|Li cell is primarily due to its higher thermodynamic stability of SLEI, thereby promoting Li⁺ transfer at NCM622|CSE interface. Besides, high-rate long cycling performance profile of NCM622|Li with B, F-CSE cycled at 1.0C is demonstrated in Figure 5e. The initial discharge capacity can reach up to 152.3 mAh g⁻¹ and maintain as high as 143.8 mAh g⁻¹ after 300 cycles with a capacity decay rate of 5.6%.

Considering that the best stability is obtained when the charge cutoff voltage is 4.3 V, the galvanostatic charge-discharge tests are also carried out with the cutoff voltage of 4.4 and 4.5 V. When the cutoff voltage increases to 4.4 V, the NCM622|Li with B, F-CSE shows an initial discharge capacity of 191 mAh g⁻¹ at 0.1C (Figure 5f), and slightly decreases to 187 mAh g⁻¹ after 100 cycles (Figure 5g). When further charged to 4.5 V, the initial discharge capacity reaches 194 mAh g⁻¹ (Figure 5f), which still retains 99.4% of the initial capacity after 50 cycles (Figure 5h). To the best of our knowledge, the high voltage electrochemical performance of NCM622|Li with B, F-CSE in this work is superior to the previously reported literatures (Table S4, Supporting Information). The improved cycling performances with 4.4 and 4.5 V cutoff voltage confirm that CSE can be matched well with high-voltage cathode by introducing B, F-donating liquid electrolyte, thus gaining an excellent interfacial thermodynamic stability, which is a consequence of the low-resistance and high-stable SLEI protection.

2.6. SLEI Formation Mechanism Analysis

To elucidate the SLEI formation mechanisms in various hybrid solid-liquid system, we simultaneously monitored their morphologies and components in full batteries cycled with various liquid electrolytes by cross-sectional SEM, corresponding EDS analyses, and XPS, respectively. As shown in Figure S11a-c in the Supporting Information, a SLEI with thickness above 10 μm is generated on the surface of F-CSE, while B, F-CSE displays a thin (1–2 μm) and smooth SLEI layer (Figure 6a),

favoring a fast Li⁺ transportation and a well-protected function of SLEI for cathode. The corresponding EDS images prove that F and B as major elements for SLEI layers are evenly distributed on the top surface of CSE (Figure 6b–d), which suggested that the SLEI may be specifically derived from the decompositions of BOB⁻ anions and fluorinated solvents.

Furthermore, XPS was applied to probe the SLEI components formed in B, F-CSE and F-CSE, respectively (Figure 6e–g; Figure S11d,e and Table S3, Supporting Information). In F-CSE system, the peaks located at ≈287.5, 288.3, 289.9, and 290.4 eV are assigned to ROCO₂Li, (CO)_n, RCO₃Li, and Li₂CO₃, respectively (Figure S11d, Supporting Information), which is common composition in alkyl carbonate electrolytes as reported in previous publications.^[42] Moreover, PEO-like polymer species (–(CH₂–CH₂–O)–; C1s: 286.5 eV) is detected in the SLEI layer on account of the existence of FEC in liquid electrolyte.^[42] According to the F1s spectra, extremely high LiF content is observed mainly due to highly fluorinated cosolvent in F-CSEs system (Figure S11e, Supporting Information).^[43] The previous studies have shown that the LiF could hinder the side reaction within the electrode|electrolyte interface.^[44] However, the overaccumulation of highly resistive LiF (10⁻⁹–10⁻¹¹ S cm⁻¹) in the interfacial layer can lead to sluggish Li⁺ transport kinetics, resulting in the large polarization and rapid capacity fading, as can be verified in Figure 5a,b,d and Figure S10a in the Supporting Information. Intriguingly, XPS confirms the relatively low LiF contents in the SLEI layer formed by the B, F-CSE (Figure 6f), in sharp contrast with that by the F-CSE. These above analyses hint that LiBOB additive can effectively regulate the LiF content in the highly fluorinated electrolytes. Moreover, in B, F-CSE system, the peak of the B-F bonding occurs at 193.6 eV in the B1s spectrum (Figure 6g) along with the peak at 686.6 eV in the F1s spectrum (Figure 6f) due to the formation of Li_xBO_yF_z ionic conducting, thereby facilitating the interfacial Li⁺ transportation.^[24,45–47] This phenomenon is also consistent with its lower *E*_a value (Figure 4f). Furthermore, NMR analyses of battery cycled only five times confirm the existence of LiDFOB intermediate products (¹¹B NMR signal at 3.06 ppm; ¹⁹F NMR signal at 151 ppm) (Figure S12, Supporting Information). Based on these, the formation mechanism of Li_xBO_yF_z could be interpreted as followed: highly fluorinated cosolvents were readily decomposed into strong nucleophile F⁻ in B, F-donating liquid electrolyte, which catalyze the decomposition reaction of the LiBOB to LiDFOB. Then, the LiDFOB further transformed into Li_xBO_yF_z ionic conducting in the presence of the large excess of F⁻. Benefiting from the advantages of organic PEO-like polymer and inorganic species LiF, the SLEI preserves the cathode with high physical and chemical compatibility, leading to enhanced interfacial stability. Simultaneously, the Li⁺ transportation dynamics at NCM622|CSE interface are also promoted, ascribing to the formation of Li_xBO_yF_z ionic conducting. As a consequence, in situ formed SLEI endows NCM622|CSE interface with low resistance and high stability, providing an excellent cycling performance and rate capability in high voltage SSBs.

To explore the protection effect of B, F-donating liquid electrolyte for high-voltage cathode in SSBs, the cycled NCM622 cathode was harvested from NCM622|Li battery, SEM, transmission electron microscopy (TEM) and corresponding

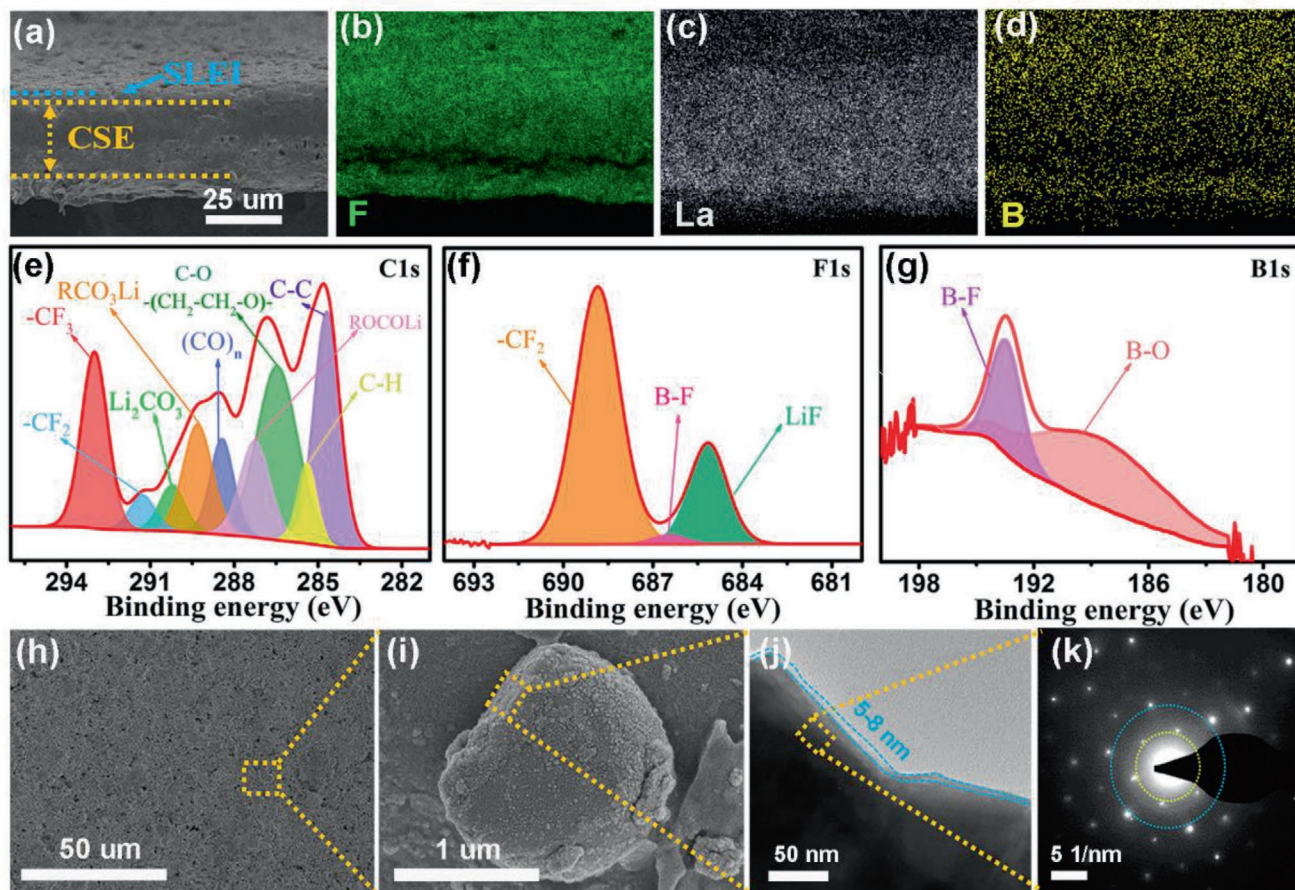


Figure 6. Characterizations of the B, F-CSE and NCM622 cathode collected from the NCM622|B, F-CSE|Li cell after 100 cycles at 0.1C. a–d) SEM image and corresponding EDS mappings of the cycled B, F-CSE. e–g) XPS spectra of C1s, F1s, and B1s regions collected from the top-surface of B, F-CSE facing with NCM622 cathode. h,i) SEM images, j) TEM image, and k) corresponding FFT image of the cycled NCM622 cathode.

diffraction patterns snapshots are shown in Figure 6h–k. After 100 cycles, the morphology of NCM622 is well preserved in NCM622|Li battery with B, F-donating liquid electrolyte defending (Figure 6h,i). Moreover, the uniform amorphous layer with 5–8 nm is covered on the particle (Figure 6j), and the corresponding FFT pattern of the near-surface region demonstrates the ordered lattice diffraction pattern, which corresponds to the crystal plane of (101) and (110) of original layered structure (Figure 6k). In contrast, obvious cracks on cathode surface and thick covering of NCM622 particle are observed in cycled NCM622|Li battery with F-CSE (Figure S11f, Supporting Information), which is further confirmed by its unsatisfactory cycling stability.

3. Conclusion

In order to achieve competitive energy density and high security, in this work, a nonflammable B, F-donating liquid electrolyte is introduced to engineer a low resistance and high stable SLEI between high-voltage cathode and ultrathin nano-LLZTO/PVDF-HFP CSE (25 μm) interface. B, F-donating liquid electrolyte shows superior chemical compatibility with CSE and

promises B, F-CSE with enhanced ionic conductivity, Li^+ transference number, and tolerated oxidation potential. Furthermore, benefiting from the existence of moderate amounts of LiF and fast ion conductor $\text{Li}_x\text{BO}_y\text{F}_z$ in SLEI formed by the decomposition reaction of BOB^- anion and highly fluorinated solvents, the NCM622|NCM622 symmetric cell renders a dramatically reduced interfacial resistance with low energy barrier for Li^+ transportation. Based on these merits, the as-assembled NCM622|Li full battery delivers superior high-voltage cycling durability (94.4% capacity retention over 300 cycles at 1.0C at 2.8–4.3 V; 97.9% capacity retention over 100 cycles at 0.1C at 2.8–4.4 V; 99.4% capacity retention over 50 cycles at 0.1C at 2.8–4.5 V) and rate capability (146 mAh g^{-1} at 2.0C). Our work proposes an innovative design concept to manipulate SLEI for stable interfacial chemistry in high-voltage SSBs, which would also be applicable to other hybrid solid–liquid systems, such as quasi-solid-state Li–S or Li– O_2 batteries.

4. Experimental Section

Preparation of CSEs: CSEs were prepared by mixing PVDF-HFP ($M_n = 400\,000\text{ g mol}^{-1}$, Sigma-Aldrich), nano-LLZTO powders (Jining Kelaitage New Energy Technology Co., Ltd.), and LiTFSI ($\geq 99.0\%$,

Sigma-Aldrich), respectively. Taking PVDF-HFP/50 wt% nano-LLZTO (CSE) as an example, 0.4 g PVDF-HFP and 0.08 g LiTFSI were dissolved in 3.0 mL 1-methyl-2-pyrrolidinone (NMP; anhydrous, >99.5%; Sigma-Aldrich), and stirred at 60 °C for 3 h to form a homogeneous solution. Then, 0.4 g nano-LLZTO powder was thoroughly dispersed into 1.0 mL acetone by repeat stirring, followed by pouring into the above mixing solution. After continually being stirred for another 24 h, the obtained light-brown solution was casted on a polytetrafluoroethylene (PTFE) mold and dried in an electricity heat drum wind drying oven at 70 °C to completely remove the NMP and acetone. Finally, the free-standing membranes were punched into pieces with diameter of 16 mm and average thickness of $25 \pm 1 \mu\text{m}$ for later use.

Preparation of Liquid Electrolyte: The B, F-donating liquid electrolyte was prepared by dissolving LiBOB (0.058 g; 0.05 m; ≥99.0%, Aladdin) and LiTFSI (1.6364 g; 0.95 m) in a mixture solvent of 1.0 mL FEC (>98%, Aladdin), 1.0 mL HFE (99%, Aladdin), and 4.0 mL PFEMC (homemade; the detailed synthetic process is given in the Supporting Information). The F-donating liquid electrolyte as a control sample was prepared by dissolving LiTFSI (1.7725 g; 1.0 m) in a mixture solvent of 1.0 mL FEC, 1.0 mL HFE, and 4.0 mL PFEMC. Conventional alkyl carbonate liquid electrolyte as a control sample was prepared by dissolving LiPF₆ (0.3038 g; 1.0 m) in a mixture solvent of 1.0 mL ethylene carbonate (EC; >98%, Aladdin) and 1.0 mL dimethyl carbonate (DMC; >98%, Aladdin). All the liquid electrolytes were prepared in an Ar-filled glovebox (H₂O, O₂ < 0.1 ppm). Quantification of water content in the resulting electrolyte solution was less than 20 ppm, determined by Karl Fischer titration.

Materials Characterization: XRDs (Smartlab, Rigaku) were used to characterize the crystallinities of the samples. The microstructures and morphologies of the samples were tested by SEM (SU8010, Hitachi) and TEM (JEM-2100F, JEOL). The elements dispersing of the specimens were characterized by energy dispersive spectrometer (Ametek). The chemical structures of samples were analyzed by FTIR spectroscopy (Nicolet 6700, Thermo Fisher Scientific). The NMR (Bruker Avancell HD 600 MHz) spectra of ¹¹B NMR and ¹⁹F NMR were conducted to determine the reaction products by using dimethyl sulfoxide-d (DMSO-d₆) as solvent. The flammable tests were proceeded by an intuitive ignition tests. XPS (ESCALab 250Xi, Al-Kα, $h\nu = 1486 \text{ eV}$, Thermo Scientific) was applied to explore the SLEI components. The binding energy was calibrated with the C1s peak at 284.6 eV. The XPS profiles were fitted using a Shirley background subtraction and a Gaussian/Lorentzian (70%/30%) peak shape.

Electrochemical Measurements: The EIS measurements of symmetric stainless steel (SS; Hefei Kejing Materials Technology Co., Ltd.)|SS cells (2.5 μL cm⁻² liquid electrolyte on both sides) were used to measure ionic conductivities in the frequency range from 10⁻¹ to 10⁶ Hz with a perturbation of 10 mV. The ionic conductivity can be calculated by formula (1) as follow

$$\sigma = \frac{L}{RS} \quad (1)$$

The R is total resistance of the electrolyte, and L and S are the thickness and surface area of electrolyte membrane, respectively. The linear sweep voltammogram tests of Li (≥99.9%, China Energy Lithium Co., Ltd.)|Super P cells (2.5 μL cm⁻² liquid electrolyte on Super P side) were performed at a scan rate of 0.1 mV s⁻¹ over the potential range of open-circuit voltage -5.5 V (vs Li⁺/Li) to evaluate the intrinsic electrochemical stable window of the electrolyte membrane. Li-ion transference number (t_{Li^+}) was measured by AC impedance and DC polarization using symmetric Li|Li cells (2.5 μL cm⁻² liquid electrolyte on both sides), and calculated by the Bruce Vincent Evans Equation (2)

$$t_{\text{Li}^+} = \frac{I_s(\Delta V - I_o R_o)}{I_o(\Delta V - I_s R_s)} \quad (2)$$

The initial and steady-state currents are defined as I_o and I_s , respectively. The initial and steady-state interfacial resistances between the electrolyte and electrodes are defined as R_o and R_s , and ΔV is the applied polarization voltage.

To investigate the electrochemical kinetic processes at the interfaces on the cathode and anode sides, EIS measurements were conducted for LiNi_{0.6}Co_{0.2}Mn_{0.2}O₂ (NCM622) symmetric cells and Li symmetric cells, separately. For NCM622|NCM622 symmetric cells fabrication, CSE was sandwiched between two NCM622 cathodes in an Ar-filled glovebox (H₂O, O₂ < 0.1 ppm) with 2.5 μL cm⁻² liquid electrolyte on both sides. For Li symmetric cells fabrication, CSEs were sandwiched between two Li|Li metal discs without liquid electrolyte at the interfaces. All EIS measurements under different temperatures were conducted in a MQ-DT(H)30F-2 high-low temperature chamber (ZhongkeMeiqi (Beijing) Technology Co., Ltd.), and the frequency ranged from 10⁻¹ to 10⁶ Hz under a voltage amplitude of 10 mV. All of the electrochemical measurements were conducted on a PARSTAT 4000 (Ametek) electrochemical workstation.

The full battery comprised NCM622 cathode, Li anode, and CSE with 2.5 μL cm⁻² liquid electrolyte on NCM622 cathode side. The cathode composition was 80 wt% NCM622, 10 wt% Super P, and 10 wt% poly(vinylidene fluoride) (PVDF, solef6020, Solvay) using NMP as dispersing solvent. The obtained slurry was scraped onto an aluminum (Al) foil collector, followed by drying at 120 °C for 12 h in vacuum. Eventually, the active material loading of NCM622 cathode was about $3.0 \pm 0.1 \text{ mg cm}^{-2}$. The assembled NCM622|Li full batteries were cycled with various current densities (1C = 180 mAh g⁻¹) at 25 °C overpotential range of 2.5–4.3 V (vs Li⁺/Li), 2.5–4.4 V (vs Li⁺/Li), and 2.5–4.5 V (vs Li⁺/Li), respectively, using a battery test system (NEWARE, China). All the batteries were allowed to rest at 50 °C for 5 h before every electrochemical cycle.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

Research data are not shared.

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